

HealthConnect Clinic

Week 4 Project Summary

Data Analytics Track

1. Specific Problem to Be Addressed

The Data Analytics track will focus on understanding appointment attendance and no-show patterns at HealthConnect Clinic. The analysis aims to identify patterns and factors that may be associated with missed appointments, which can affect appointment-slot utilisation and overall clinic efficiency.

The central analytical focus is to explore how appointment characteristics, previous appointment behaviour, reminders, accessibility factors and patient demographics may be associated with appointment outcomes.

2. Resources Used

- HealthConnect Appointment Dataset containing 5,000 appointment records and 18 variables.
- HealthConnect Data Dictionary, used to understand the definitions and expected data types of each variable.
- Power BI and Power Query, used to review the dataset structure and conduct the initial data-quality assessment.
- The Week 4 project guide, used to define the required analytical tasks and deliverables.

3. Key Observations from Week 4

- The dataset contains 5,000 appointment records and 18 variables covering patient demographics, appointment details, booking history, reminders, accessibility factors, waiting time and appointment outcomes.
- No duplicate appointment records were identified.
- All variables had data types consistent with the Data Dictionary.
- distance_to_clinic_km contains 90 missing values (2%).
- waiting_time_minutes contains 60 missing values (1%).
- No inconsistent or invalid values were identified during the initial review.
- The initial review of key numeric variables did not identify any obvious unusual values or outliers.
- Several variables were identified as relevant for investigating attendance and no-show patterns, particularly appointment outcome, previous no-shows, booking lead time, reminders, appointment characteristics, distance to the clinic and waiting time.

4. Proposed Approach

1. Review and address the identified missing values where necessary.
2. Analyse appointment_outcome to establish baseline attendance and no-show rates.
3. Investigate no-show patterns across patient demographics and appointment characteristics.
4. Examine the relationship between previous no-shows and future appointment outcomes.
5. Explore whether reminders and reminder channels are associated with attendance.
6. Investigate the potential influence of booking lead time, distance to the clinic and waiting time.
7. Develop and refine relevant KPIs and visualisations based on the findings.
8. Identify key patterns and opportunities that may support efforts to reduce missed appointments.

5. Key Considerations Affecting the Project

- The analysis can identify patterns and associations, but the available data alone may not establish direct causation.
- The interpretation of attendance and no-show KPIs will depend on the exact categories contained in appointment_outcome.
- Other factors that may influence appointment attendance may not be represented in the available dataset.
- Consistent definitions and calculation methods will be required when developing KPIs and future dashboard measures.
- The missing values in distance_to_clinic_km and waiting_time_minutes may affect analyses involving these variables.

6. Proposed Focus for Week 5

- Calculating the overall appointment attendance and no-show rates.
- Analysing no-show patterns across demographic and appointment-related variables.
- Investigating the relationship between previous no-shows and appointment outcomes.
- Exploring reminder effectiveness and differences between reminder channels.
- Analysing the potential relationship between accessibility, waiting time and no-show patterns.
- Refining the proposed KPIs based on the findings.
- Identifying the most important factors for further investigation and future dashboard development.